



Tribhuvan University

Faculty of Humanities and Social Science

Project on

Job Portal

Submitted to

Department of Computer Application

College Name

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

Your name Tu ID

July 2025

Under Supervision of

Supervisor name



Tribhuvan University
Faculty of Humanities and Social Science
Pascal National College

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by Rajeshan Maharjan entitled “**Job Portal**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

Basanta Chapagain

SUPERVISOR

Department of Computer Application

Pascal National College, Satdobato, Lalitpur



Tribhuvan University

Faculty of Humanities and Social Science

Pascal National College

LETTER OF APPROVAL

This is to certify that this project prepared by Rajeshan Maharjan entitled "Job Portal" in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

Basanta Chapagain Department of Computer Application Pascal National College Satdobato, Lalitpur	Suresh Thapa BCA Coordinator Department of Computer Application Pascal National College Satdobato, Lalitpur
Basanta Chapagain Internal Examiner	 (External Examiner)

ABSTRACT

This project presents the design and implementation of a Job Portal web application developed as part of the Bachelor in Computer Application (BCA) curriculum. The system aims to address the limitations of traditional recruitment methods, which are often time-consuming, fragmented, and inaccessible to many job seekers and employers in Nepal. The main objective is to provide a centralized, user-friendly, and secure digital platform where employers can post vacancies and manage applications, while job seekers can create profiles, search for jobs, and apply efficiently.

The system was developed using a three-tier architecture with React.js for the frontend, Node.js for the backend, and MongoDB as the database. Agile methodology was adopted to manage iterative development, while system analysis and design were carried out using ER diagrams, DFDs, and flowcharts. For intelligent job matching, text preprocessing, tokenization, stopword removal, and Jaccard similarity were applied to compare resumes with job descriptions and generate recommendations.

The completed portal successfully streamlines recruitment, ensures secure user authentication, and provides efficient job–applicant matching. This demonstrates its potential to improve hiring processes and reduce manual effort in the Nepali context.

Keywords: Job Portal, Web Application, Jaccard Similarity, NLP, Recruitment System, MongoDB, Node.js, Tokenization, Resume Matching, Job Recommendation

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude and appreciation to all those who have supported and guided us throughout the development of the Job Portal.

First and foremost, we would like to extend our deepest thanks to our project supervisor, **Basanta Chapagain** and our coordinator **Suresh Thapa**, for their invaluable guidance, expertise, and constant support. Their mentorship and constructive feedback have been instrumental in shaping the project and pushing us to deliver our best.

Lastly, we would like to acknowledge the countless individuals who have shared their knowledge and expertise, either through discussions, feedback, or research materials. Your contributions have greatly enriched our understanding and helped shape the final outcome of this project.

Sincerely,

Rajeshan Maharjan

LIST OF ABBREVIATIONS

API	Application Programming Interface
DFD	Data Flow Diagram
ERD	Entity Relation Diagram
HTML	Hyper Text Markup Language
JS	JavaScript
UI	User Interface
VS Code	Visual Studio Code

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The Job portal web application is a comprehensive online platform that bridges the gap between job seekers and employers.[1] It provides users with a digital space to create and manage profiles, search for job opportunities, and submit applications seamlessly.[2] On the employer side, the portal enables companies to post job openings, view applicant details, and manage hiring workflows efficiently. The Job Portal system is designed to enhance the recruitment experience by providing intuitive user interfaces, powerful search and filter capabilities, and secure data management.[3] With increasing digitization in recruitment, platforms like this play a vital role in streamlining employment connections.

1.2 Problem Statement

Many Traditional recruitment methods often involve time-consuming manual processes, such as paper-based applications, walk-in interviews, and limited reach for both employers and job seekers. Furthermore, there's a lack of centralized platforms in the Nepali context that offer secure, efficient, and user-friendly solutions tailored for local use.

Thus, there is a need for a system that can:

- Connect job seekers with employers
- Provide an intuitive platform to manage job applications
- Ensure data privacy and security during the recruitment process.

1.3 Objectives

The objectives of this project are follows as:

- To develop a dynamic web-based job portal system
- To allow job seekers to register, search and apply for jobs
- To enable employers to post job listings and manage application
- To implement secure login, job filtering, and application tracking

1.4.1 Scopes

- User account creation for both employers and job seekers
- Job posting, search, application functionality.

1.4.2 Limitations

- No AI/ML-Based recommendations
- Video interview integration
- Create a multiple profile based on the skill
- Application Analytics for employers

1.5 Report Organization

Chapter 1: Introduction

It includes the general overview of the system and project as a whole. It includes the problem statement, Objective and Scope/ Limitations of the project and the system being developed.

Chapter 2: Background Study and Literature Review

It includes the study of the current scenario/environment the system will be deployed into. It includes the study of the current trends, preferences of people, the existing systems, areas of improvement among others.

Chapter 3: System Analysis and Design

It includes the requirement and feasibility analysis of the system that can be generated through the studies presented in the previous two chapters. It will also include the Development Methodology, Flowchart, ER and DFD for the system which specifies the workflow, entities, attributes and their relationships. It includes the design of the database, forms and interface of the system. It also includes the implementation details of the selected methodology and the details of the algorithm used.

Chapter 4: Implementation and Testing

It includes the details of the different design and development tools used and the implementation details of the modules presented in the form of code snippets of functions, classes. It also includes the testing of the system with different test cases as per the requirement.

Chapter 6: Conclusion and Future Recommendations

It includes the summary of the system and the project as a whole. It also includes the possibilities/aspects which the system can implement in the future.

CHAPTER 2

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Job portals are online platforms designed to connect job seekers with employers. These portals allow users to browse job listings, submit applications, and maintain resumes online. Employers, in turn, can post job vacancies, search through applicant profiles, and streamline the hiring process. Job portals have become an essential component of modern recruitment systems, replacing many traditional methods such as newspaper advertisements and walk-in interviews[4].

A well-designed job portal not only simplifies the job search and hiring process but also offers features like filtering jobs based on experience, location, job type, and more. With the rise of digital platforms in Nepal and globally, job portals have gained increasing popularity due to their accessibility, time efficiency, and broader reach[5].

Globally, job portals such as LinkedIn, Indeed, and Glassdoor have transformed the way individuals search for employment opportunities and how organizations attract and manage talent. These platforms integrate features like job search filters, resume uploads, application tracking, and skill-based recommendations, enabling more efficient matching between candidates and employers[6].

In the context of Nepal, the demand for effective recruitment platforms has grown rapidly due to the increasing number of graduates, rising use of digital technology, and the need for skilled manpower across industries. Local portals such as MeroJob and Kumari Job have played an important role in digitizing recruitment [7]. However, challenges such as limited personalization, lack of AI-based recommendations, and data security still exist [8].

Therefore, the development of a dedicated Job Portal system not only addresses these limitations but also provides a dynamic, user-friendly, and secure platform tailored to the needs of both employers and job seekers in the Nepali job market[9].

2.2 Literature Review

The authors in a research paper titled “Online Recruitment Platforms and Employment Dynamics” explained how digital transformation in hiring has reduced the time-to-hire by 40–60% compared to traditional methods. Online portals help companies expand their talent pool and improve access for applicants, especially in rural or underrepresented areas. The paper emphasized the importance of mobile-optimized, user-friendly interfaces for job portals.[10]

In the journal “A Comparative Study on Traditional vs. Online Recruitment”, researchers found that online job portals offer cost-effective, wider-reaching, and more efficient solutions. Traditional hiring methods suffer from geographical limitations, slower communication, and higher overhead. The study highlights how AI and filtering mechanisms used in modern portals improve match accuracy between candidates and job roles.[11]

A case study on “Digital Employment Services in South Asia” emphasized the role of job portals in bridging the employment gap in developing countries like Nepal, India, and Bangladesh. It pointed out the need for localized platforms that cater to domestic job markets, support multiple languages, and provide tools for both skilled and unskilled labor segments.[12]

In the book “Modern Web Development”, the authors discuss the critical components of building scalable job portals, including database optimization, RESTful APIs, frontend-backend integration, and security practices. They emphasize the importance of a modular structure and clean user experience for the success of such platforms.[13]

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

Systems analysis is the process of observing systems for troubleshooting or development purposes. It is applied to information technology, where computer-based systems require defined analysis according to their makeup and design.

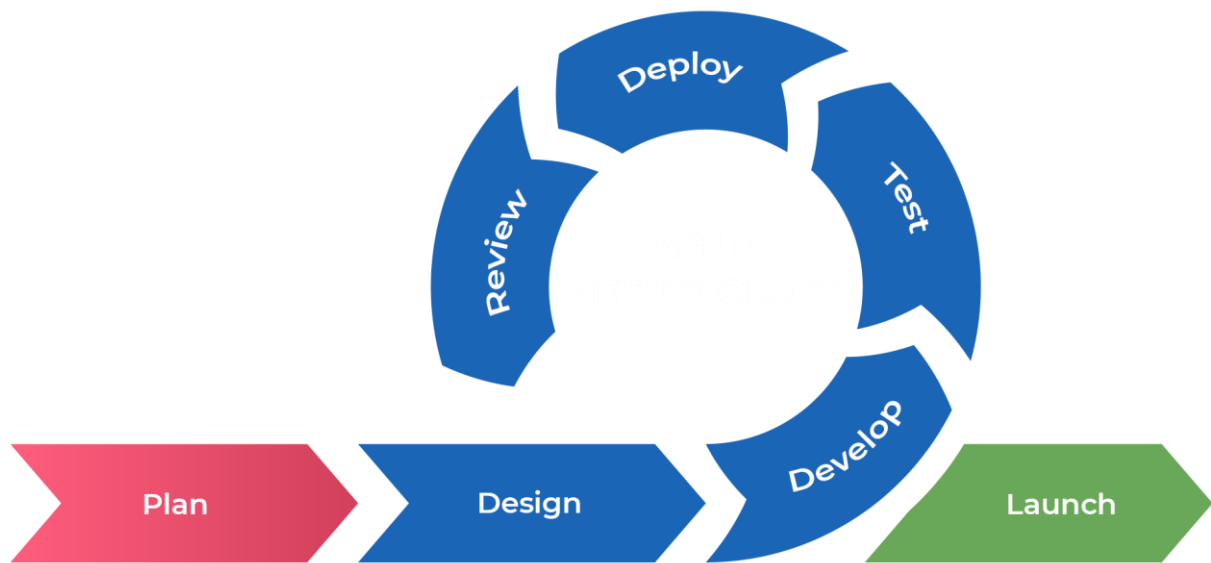


Figure 3. 1 Agile Model for Job Portal Web Application

3.1.1 Requirement Analysis

This web application needs to fulfill the given functional and non-functional requirements:

I. Functional Requirement:

- The application should be able to register and log in valid users.
- The application should be able to allow the employer to post job listings
- The application should be able to allow the job seekers to apply for jobs

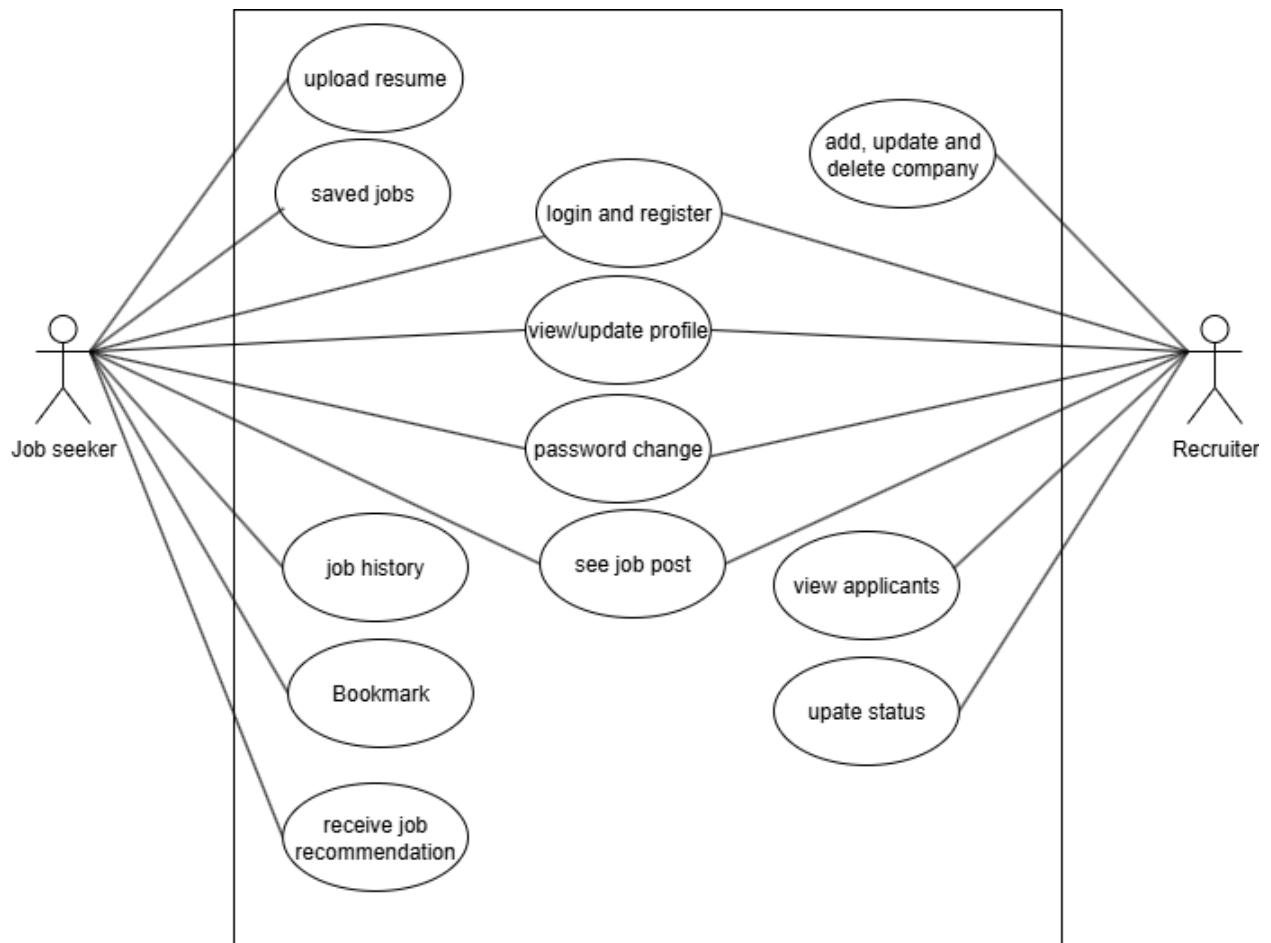


Figure 3. 2 Use-case diagram of job Portal

II. Non-Functional Requirement:

- The application should be secure and protect user data.
- The user interface should be intuitive and easy to use.
- The system should be fast and responsive.
- The application should support multiple devices (responsive design).

3.1.2 Feasibility Analysis

A feasibility study, also known as feasibility analysis, is an analysis of the viability of an idea. It describes a preliminary study undertaken to determine and document a project's viability. The results of this analysis are used in making the decision whether to proceed with the project or not. In short, a feasibility analysis evaluates the project's potential for success, following feasibility analysis was performed prior to working on the project:

I. Technical Feasibility Study

This project can be considered technically feasible as it deals with the secure storage of users notes and it can be easily implemented using the available technologies.

ii. Operational Feasibility Study

This project can be a user-friendly application with no extra knowledge required for its operation. It would be as simple as saving some sort of important data.

iii. Economic Feasibility Study

There is no much expense in developing this project as the technologies required for it are mostly open-source and easily available to use.

iv. Schedule Feasibility Study

The project expands over a period of 87 days with each tasks being carried out step by step following the system development model. It is completed within scheduled time and do not exceed the scheduled time.

Task Name	Duration
Research and Planning	10 Days
Requirement Gathering	8 Days
System Design	15 Days
System Development	40 Days
Review	25 Days
Testing	10 Days

Table 3. 1 Gantt chart table for Job portal

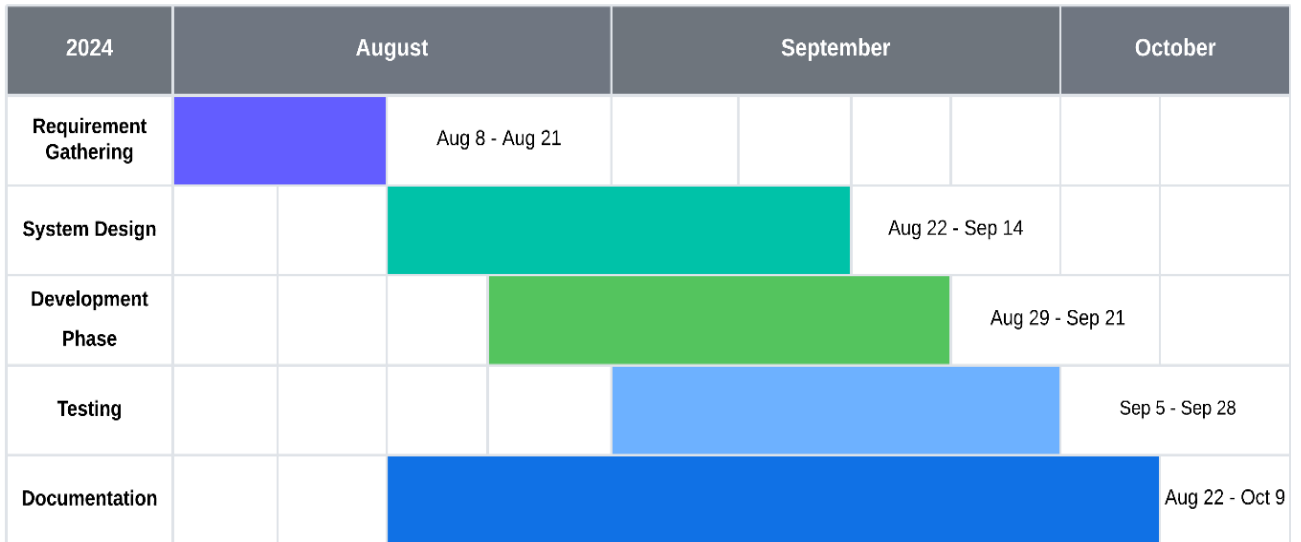


Figure 3. 3 Gantt chart of schedule timeline

3.1.3 Data Modelling

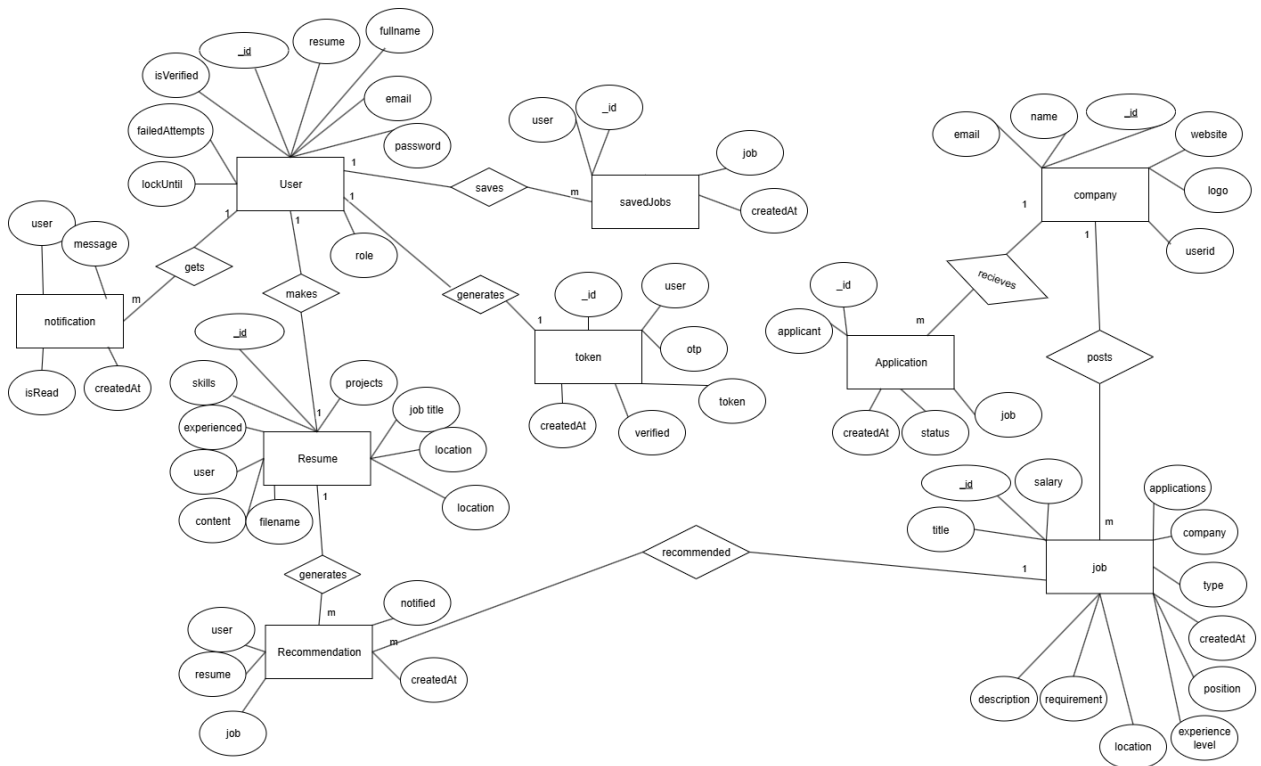


Figure 3. 4 Entity-relation diagram of job portal

In the above ER-Diagram for the Job Portal web application, we have multiple entities with their own attributes. The entity User has attributes such as `_id`, `fullname`, `email`, `password`, `role`, `resume`, `isVerified`, `failedAttempts`, and `lockUntil`. Similarly, the entity Job has different attributes such as `_id`, `title`, `description`, `requirement`, `location`, `type`, `position`, `experience level`, `salary`, `createdAt`, and `applications`. The entity Application includes attributes like `_id`,

applicant, job, status, and createdAt, which is used to store application details of the user for a particular job.

Likewise, the entity Resume has attributes such as _id, skills, projects, experienced, filename, content, job title, and location and is associated with each user profile. The entity Company consists of attributes _id, name, email, website, logo, and userId and represents the company that posts jobs. The Token entity contains _id, otp, token, createdAt, and verified used for user verification purposes such as account confirmation or password reset.

The entity Notification includes _id, user, message, isRead, and createdAt, which helps to notify users about system events. Similarly, the SavedJobs entity maintains the record of saved job listings by users and includes user, job, and createdAt. Finally, the Recommendation entity includes attributes such as _id, user, resume, job, notified, and createdAt, which helps in providing personalized job suggestions to users based on their resume and activity.

3.1.4 Process Modelling(DFD)

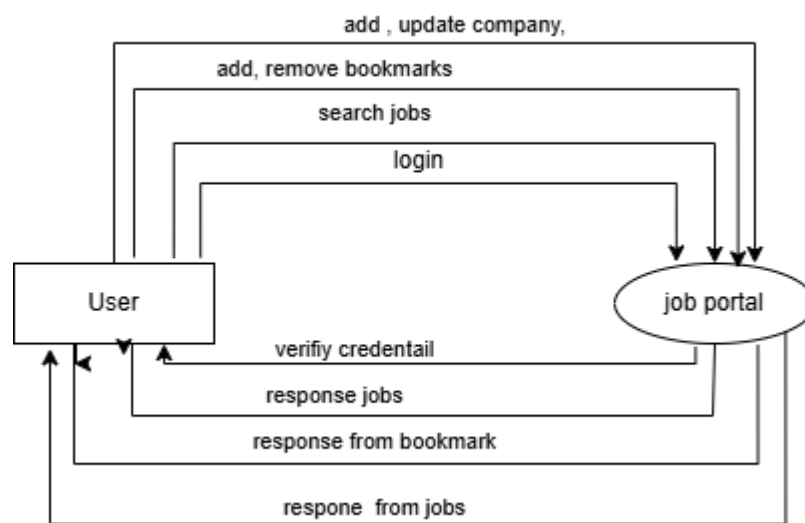


Figure 3. 5 Level 0 DFD of Job Portal

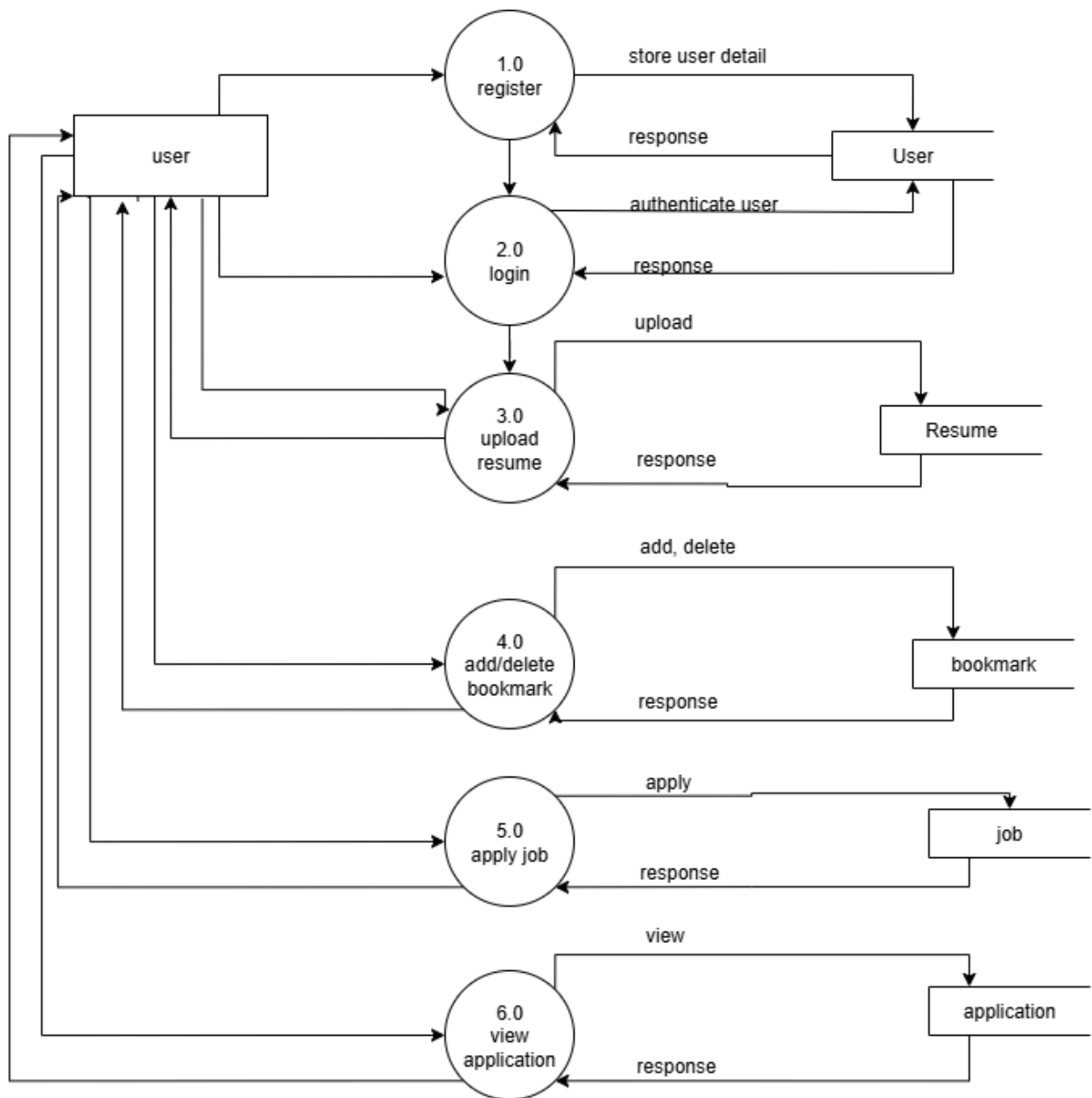


Figure 3. 6 Level1 DFD of job portal

3.2 System Design

To implement the different functional requirements of the system in graphical form, different design diagrams of the system have been prepared which are as follows:

3.2.1 Architectural Design

For this system, three tier architecture is implemented which includes user interface, web server and database. In architectural design, basic structure of the system is shown:

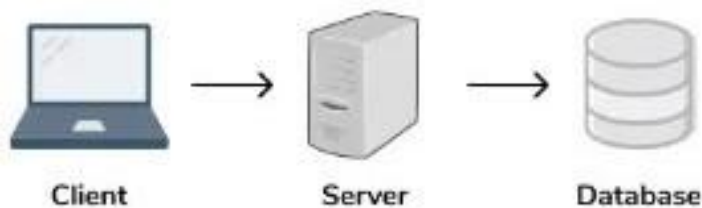


Figure 3. 7: Architecture design of Job Portal

3.2.2 System Flowchart

Flowchart is the pictorial representation of the program logic of the system. The below flowchart represents the system processes of Job Portal web application:

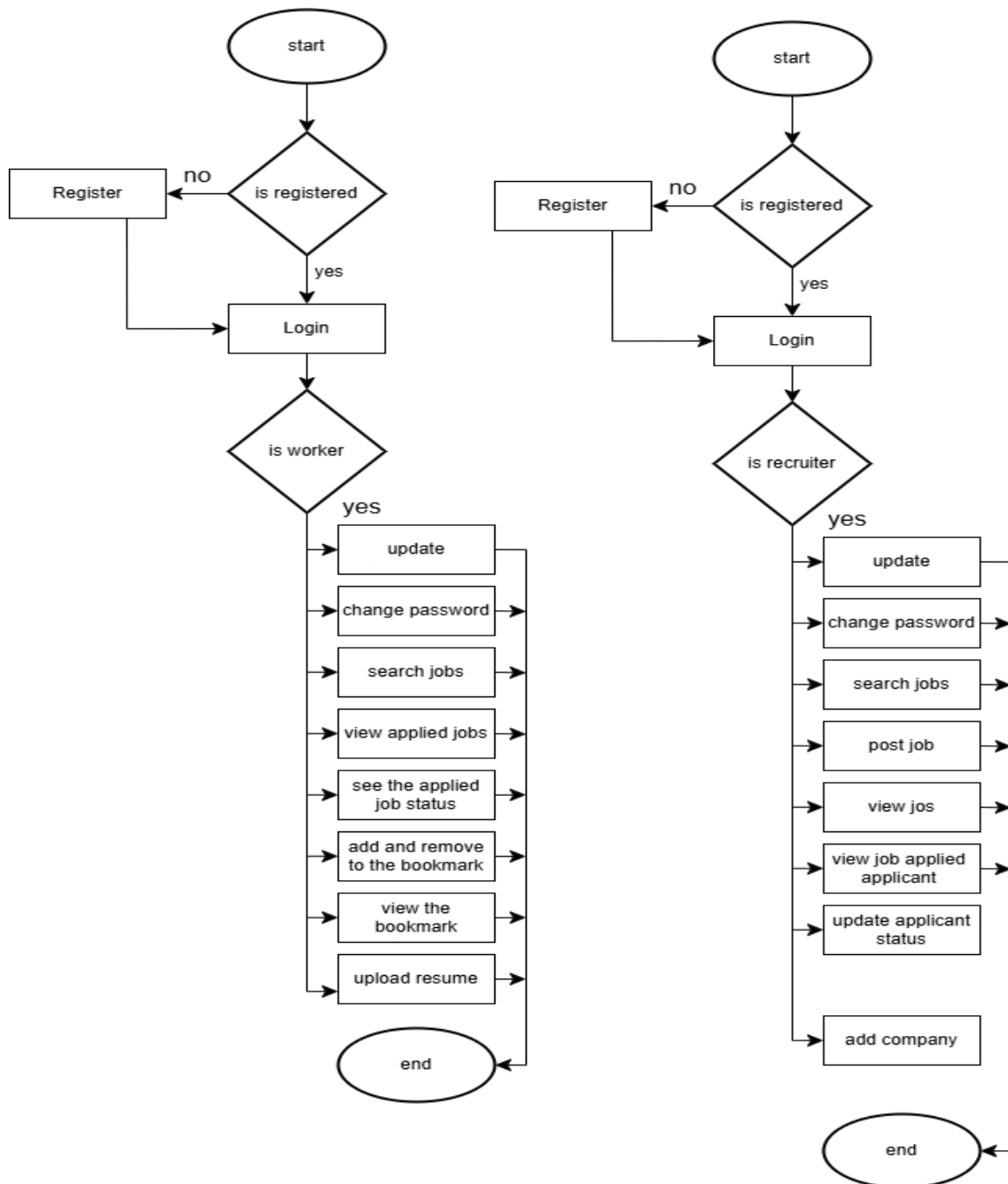


Figure 3. 8 System flowchart of job portal

3.2.3 Database Schema Design

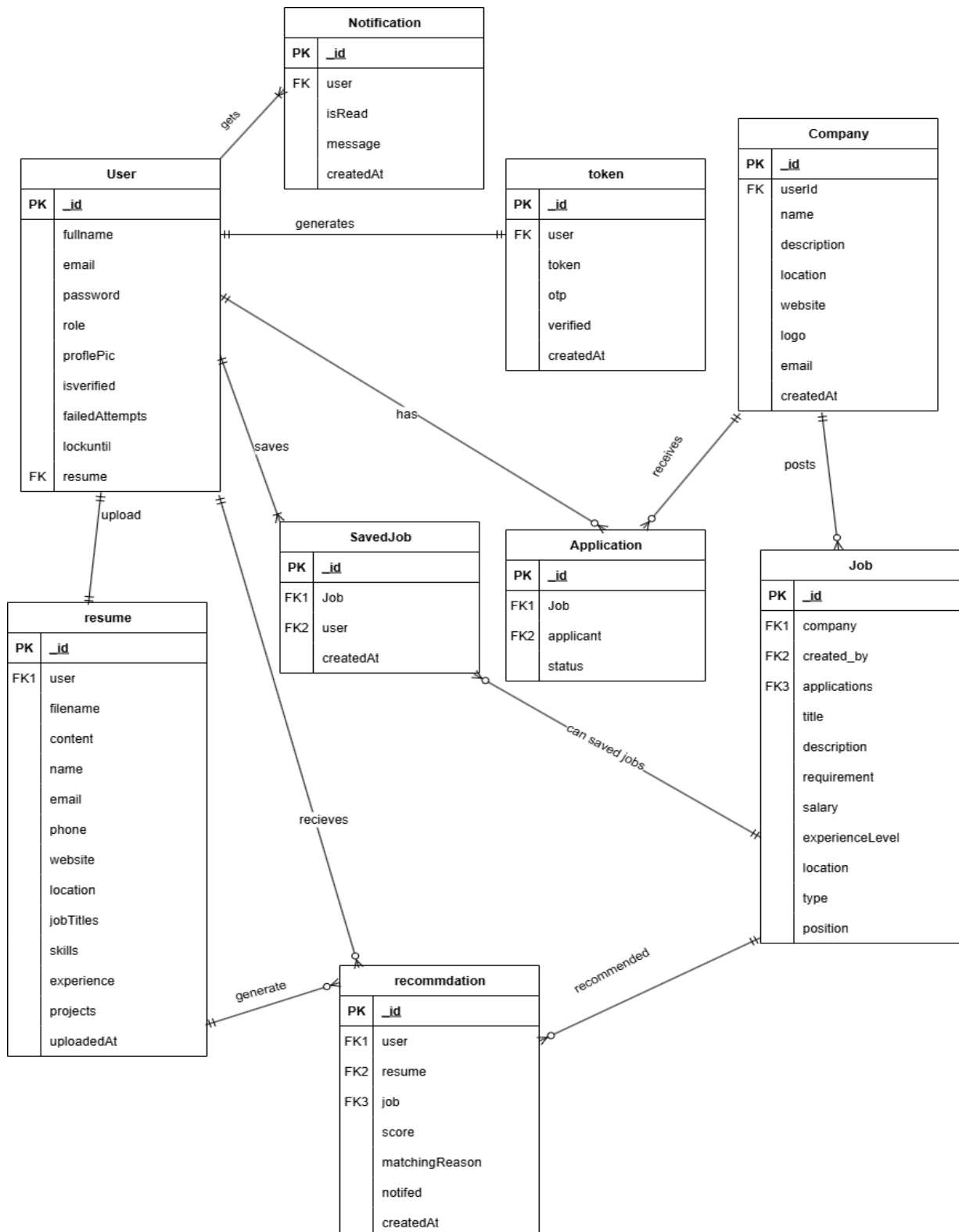


Figure 3. 9 Database schema design of job portal

The database schema includes several key tables, primarily: USER, JOB, APPLICATION, RESUME, and COMPANY. The USER table stores details such as _id, fullname, email,

password, role, and status fields like isVerified, failedAttempts, and lockUntil. Each user can be a job seeker or employer, and may have an associated RESUME and multiple APPLICATIONS, SAVED JOBS, and NOTIFICATIONS. The JOB table includes attributes like _id, title, description, requirement, location, type, position, experience level, salary, and createdAt. Each job is posted by a COMPANY, creating a one-to-many relationship where one company can post many jobs. The COMPANY table contains _id, name, email, website, logo, and userId, linking to the USER table. The APPLICATION table models job applications and connects USER and JOB in a many-to-many relationship. It includes _id, applicant, job, status, and createdAt.

The RESUME table holds a user's skills, projects, experience status, job title preference, and resume file metadata like filename and content. The RECOMMENDATION table relates users, jobs, and resumes to generate personalized job matches, while SAVEDJOBS logs which users have bookmarked which job listings.

3.2.4 User Interface Design

User interface is one of the most important parts of the system as it determines how easy it is for a new user using the system to understand the different components listed and navigate through them in order to achieve the intended goal of using the system. The interface of our system will be presented according to the following representations:

Welcome Back

Recruiter worker

Email

password

forget password

Submit

Don't have an account ? Register

Figure 3. 10 interface design for the login

Register for free

Recruiter

worker

username

Email

phone

password

Submit

have an account ? Login

Figure 3. 11 Register interface design for register page

logo

Home About jobs

Find your get
your Dream
job with us

good life with a good company. start explore
thousands of jobs in one place

Figure 3. 12 Home interface design for the job portal

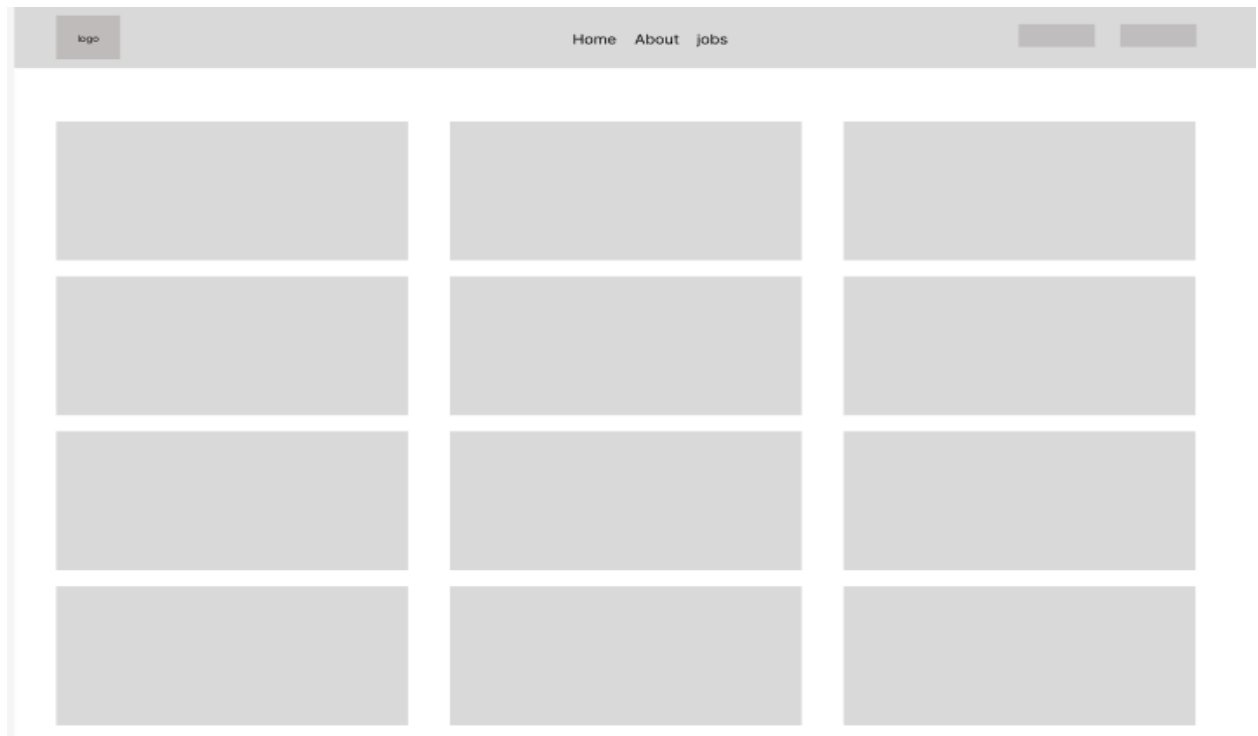


Figure 3. 13 jobs interface design for the job portal

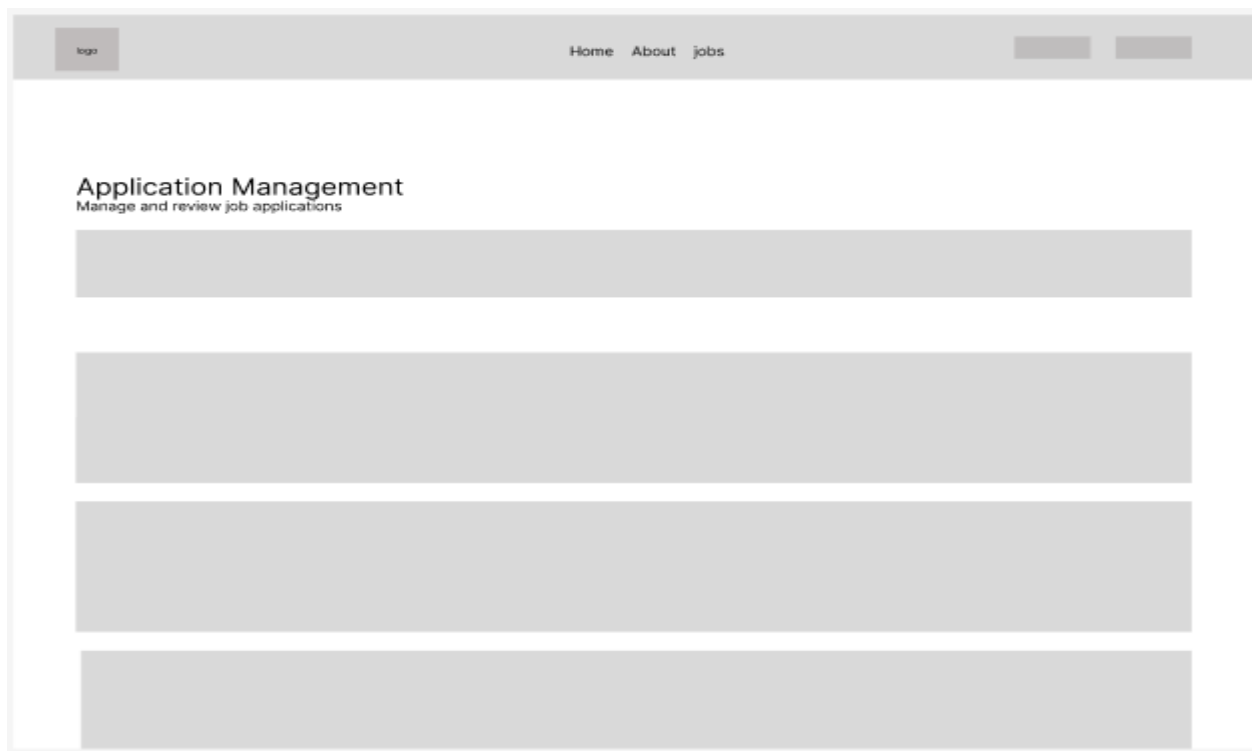


Figure 3. 14 applicant interface design for the job portal

3.3 Description of Algorithm

3.3.1 Tokenization Algorithm

The resume content and job description are first converted to lowercase to maintain consistency. The text is then tokenized, meaning it is split into individual words (tokens) by removing punctuation and special symbols. A stopwords list (containing common words like “is”, “the”, “and”) is applied to remove unimportant terms, leaving only meaningful keywords.

Jaccard Similarity Calculation:

Let the set of cleaned tokens from the resume be R, and from the job description be J. The algorithm calculates the Jaccard similarity using the formula:

$$\text{Jaccard}(R,J) = |R \cap J| / |R \cup J|$$

This measures how similar the resume is to the job description based on overlapping keywords.

Ranking & Recommendation:

Each job listing receives a similarity score based on the above formula. Jobs are then ranked in descending order of similarity, and the top matches are stored in the Recommendation collection for the user. These recommendations can then be displayed or sent as notifications.

3.3.2 Jaccard Similarity Algorithm

The Jaccard Similarity (or Jaccard Index) is a measure of similarity between two sets, commonly used in text mining, recommendation systems, and data clustering. It compares the size of the intersection of the sets to the size of their union.

Mathematical Definition

Given two sets A and B, the Jaccard Similarity is calculated as:

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

$$J(A,B) =$$

$$\frac{|A \cap B|}{|A \cup B|}$$

$$\frac{|A \cap B|}{|A \cup B|}$$

- 0 → No overlap between sets.
- 1 → Sets are identical.

Algorithm Steps

1. Convert Inputs to Sets

- For text data, tokenize and remove duplicates (e.g., using word-level or character-level tokens).

2. Compute Intersection & Union

- Intersection ($A \cap B$): Elements common to both sets.
- Union ($A \cup B$): All unique elements from both sets.

3. Calculate Similarity

Divide intersection size by union size

Example

Input:

- Text 1: "apple banana orange"
- Text 2: "banana peach apple"

Step 1: Tokenize & Convert to Sets

- $A = \{\text{"apple"}, \text{"banana"}, \text{"orange"}\}$
- $B = \{\text{"banana"}, \text{"peach"}, \text{"apple"}\}$

Step 2: Compute Intersection & Union

- $A \cap B = \{\text{"apple"}, \text{"banana"}\} \rightarrow |A \cap B| = 2$
- $A \cup B = \{\text{"apple"}, \text{"banana"}, \text{"orange"}, \text{"peach"}\} \rightarrow |A \cup B| = 4$

Step 3: Calculate Jaccard Similarity

$$J(A,B)=\frac{2}{4}=0.5$$

$$J(A,B)=$$

$$\frac{2}{4}$$

$$=0.5$$

Applications

- Document Similarity: Compare articles, tweets, or search results.
- Recommendation Systems: Find users with similar preferences.
- Duplicate Detection: Identify near-identical records in datasets.

Variants

- Jaccard Distance: $1 - J(A, B)$ (measures dissimilarity).
- Weighted Jaccard: For multisets (counts of elements matter).

3.3.3 Stopword Removal

These stopwords are common words that do not contribute significantly to the meaning of a sentence, such as “is”, “the”, “and”, “in”.

In this phase:

- A predefined list of stopwords is used (stopwords.js in backend code).
- Tokens that match any stopwords are removed from the list
- This ensures that only meaningful keywords remain for similarity calculation
- Example:
Tokens before removal: ["experienced", "web", "developer", "skilled", "in", "javascript", "and", "node", "js"]
Tokens after removal: ["experienced", "web", "developer", "skilled", "javascript", "node", "js"]

3.3.4 Ranking and Recommendation

- This similarity is calculated for each job
- Jobs are ranked from highest to lowest similarity score
- The top matches are stored in the recommendation collection and displayed to the user

CHAPTER 4

IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

Below are the tools that were used to develop and complete this project:

Front End Tools

A frontend-end application is one that application & users interact with directly. It must have a graphical user interface that assists users that are not from IT background. It must be scalable and flexible. The frontend tools that I have used in making this web application

1. React.js

React is a JavaScript library for building user interfaces (UIs). It lets developers create reusable components (like buttons, forms, or entire pages) that update dynamically without reloading the page.

Database

1.MongoDB(Mongoose)

MongoDB is a NoSQL database that stores data in flexible, JSON-like documents (called BSON). Unlike traditional SQL databases (e.g., MySQL), it doesn't use tables—instead, data is organized in collections (like folders) and documents (like files).[14]

Backend Tools

Node.js is an open-source, cross-platform JavaScript runtime environment that allows developers to run JavaScript on the server side. It is built on the Google Chrome V8 engine and is designed to be lightweight, efficient, and scalable for building fast network applications.

Documentation tools

Microsoft word is used to create the required documents for the projects. It is a word processing application developed by Microsoft. It is most widely used tools for creating, editing, formatting and sharing text-based documents. It supports various types, including .docx and .pdf and provides a range of features for professional and academic use

4.1.2 Implementation Details of Modules

1. login module

A user can register to the website providing the information such as name, email and password. Once they are registered to the website, then they can login to the website providing the valid credentials that is, a valid email and a valid password.

Code for user registration model:

```
export const register = async (req, res) => {
  try {
    const { fullname, email, password, role } = req.body;

    const user = await User.findOne({ email });

    if (user) {
      throw new Error("Already user exists.");
    }

    if (!email) {
      throw new Error("Please provide email");
    }

    if (!password) {
      throw new Error("Please provide password");
    }

    if (password.length < 7) {
      throw new Error("password mus contain atleast 7 char");
    }

    if (!fullname) {
      throw new Error("Please provide fullname");
    }

    if (!role) {
      throw new Error("Please provide role");
    }
  }
}
```

```

}

const salt = bcrypt.genSaltSync(10);
const hashPassword = await bcrypt.hashSync(password, salt);

if (!hashPassword) {
  throw new Error("Something is wrong");
}

const payload = {
  ...req.body,
  password: hashPassword,
};

const userData = new User(payload);
const saveUser = await userData.save();
saveUser.password = undefined;

res.status(201).json({
  data: saveUser,
  success: true,
  error: false,
  message: "User created Successfully!",
});
} catch (err) {
  res.json({
    message: err.message || err,
    error: true,
    success: false,
  });
}

```



```
});  
}  
};
```

2. Text extraction from pdf/word and calculate similarity

```
export async function fileExists(filePath) {  
  try {  
    await fs.access(filePath);  
    return true;  
  } catch {  
    return false;  
  }  
}  
  
// PDF Extractor  
export const extractTextFromPdf = async (filePath) => {  
  try {  
    const pdfExtract = new PDFExtract();  
    const data = await pdfExtract.extract(filePath, {});  
    return data.pages.map(page => page.content.map(item => item.str).join(' ')).join('\n');  
  } catch (error) {  
    console.error("PDF extraction error:", error);  
    return null;  
  }  
};  
  
// DOCX Extractor  
export const extractTextFromDocx = async (filePath) => {
```

```

try {
  const { value } = await mammoth.extractRawText({ path: filePath });
  return value;
} catch (error) {
  console.error("DOCX extraction error:", error);
  return null;
}
};

// Resume Detail Extractor
export const extractDetail = (text, filename) => {
  if (!text) return {};

  const lines = text.split('\n').map(line => line.trim()).filter(line => line.length > 0);
  const cleanText = text.replace(/\r?\n|\r/g, ' ').trim();

  // 1. Name (first line or capitalized name-like pattern)
  let name = "Unknown";

  if (lines.length > 0 && /^[A-Z\s]{5,}$/.test(lines[0])) {
    name = lines[0].split(' ').map(word => word.charAt(0).toUpperCase() +
word.slice(1).toLowerCase()).join(' ');
  } else {
    const nameMatch = cleanText.match(/\b[A-Z][a-z]+\s[A-Z][a-z]+/);
    if (nameMatch) name = nameMatch[0];
  }

  // 2. Email
  const emailMatch = cleanText.match(/\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Z|a-
z]{2,}\b/);

```

```

const email = emailMatch ? emailMatch[0] : "Not Found";

// 3. Website
const websiteRegex = /(https?:\V\[\^s\]|www\.\[\^s\]|)/i;
const websitematch = cleanText.match(websiteRegex);
const website = websitematch ? websitematch[0].trim() : ";

// 4. Phone
const phoneMatch = cleanText.match(/(\+?\d{1,3}[-.\s])?(?(\d{2,4})?[-.\s])?\d{3,4}[-.\s]?d{4}/);
const phone = phoneMatch ? phoneMatch[0] : "Not Found";

// 5. Location (try from address or known cities)
const locationMatch =
cleanText.match(/\b(Siddhipur|Lalitpur|Kathmandu|Bhaktapur|Pokhara|Butwal|Nepalgunj
)\b/i);
const location = locationMatch ? locationMatch[0] : "Not Found";

// 6. Skills
const skillRegex =
\b(HTML5?|CSS3?|JavaScript|React|Vue\.js?|Angular|Bootstrap|Tailwind|GitHub|GitLab
|GraphQL|TypeScript|Webpack|NPM|Babel|Jest|Figma|Adobe
XD|Sketch|Cypress|Jest|REST|Responsive Design)\b/gi;
const skills = [...new Set([...cleanText.matchAll(skillRegex)].map(m => m[0]))];

// 7. Job Titles
const jobTitleRegex = \b(Frontend|Backend|Full Stack|UI/UX|Web) Developer\b/gi;
const jobTitles = [...new Set([...cleanText.matchAll(jobTitleRegex)].map(m => m[0]))];

// 8. Multiple Experience Dates
const experienceRegex = \b(?:Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)[a-

```

```

z]*[\s.-]*\d{4}\s*[-—to]+\s*(?:Present|w{3,9}\s\d{4}|\d{4})/gi;

const experience = [...cleanText.matchAll(experienceRegex)].map(m => m[0]);

// objective
let objective = "";

const objectiveStart = lines.findIndex(line => /^Objective$/i.test(line));
if (objectiveStart !== -1) {
  for (let i = objectiveStart + 1; i < lines.length; i++) {
    const line = lines[i];

    if (/^(Experience|Education|Skills|Projects|Certifications)$/i.test(line)) break;

    objective += line + " ";
  }
  objective = objective.trim();
}

// 9.project
const projectStart = lines.findIndex(line => /^Projects$/i.test(line));
let rawProjects = [];
if (projectStart !== -1) {
  for (let i = projectStart + 1; i < lines.length; i++) {
    if (/^(Experience|Education|Skills|Certifications|Objective)$/i.test(lines[i])) break;
    if (lines[i].length > 10) rawProjects.push(lines[i]);
  }
}

const projects = rawProjects.map(project => {
  const [title] = project.split(/(<=\\)|(<=\\.)/);
  return title.trim();
});

```

```

return {
  filename,
  content:text,
  objective,
  name,
  email,
  phone,
  location,
  jobTitles,
  skills,
  website,
  experience,
  projects,
};
};

// Simple Jaccard Similarity
export const calculateSimilarity = (text1, text2) => {
  if (!text1 || !text2) return 0;

  // Convert to lowercase and tokenize by splitting on non-word characters
  const tokens1 = text1.toLowerCase().split(/\W+/).filter(token => token.length > 2);
  const tokens2 = text2.toLowerCase().split(/\W+/).filter(token => token.length > 2);

  // Create sets of unique tokens
  const uniqueTokens1 = new Set(tokens1);
  const uniqueTokens2 = new Set(tokens2);

  // Find intersection

```

```

const intersection = new Set([...uniqueTokens1].filter(token =>
uniqueTokens2.has(token)));

// Calculate Jaccard similarity (intersection over union)
const union = new Set([...uniqueTokens1, ...uniqueTokens2]);

return intersection.size / union.size;

};

```

4.2 Testing

4.2.1 Test cases for unit testing

S N	Test Name	Input	Expected Output	Actual Output	Test Result
1	Register to Application	http://localhost:8081/ap i/v1/user/register	Register page	Reigster page	Pass
2	Enter invalid information	username = 'demo, password = 'Demo@123',	Enter email	Please enter email.	Pass
3	Enter valid information	username = 'Demo, password = 'Demo@123', email = demo@gmail.com	User registered successfully	User registered successfull y.	Pass

Table 4. 1 Test case for user register

S N	Test Name	Input	Expected Output	Actual Output	Test Result
--------	-----------	-------	--------------------	------------------	----------------

1	Login to Application	http://localhost:8081/api/v1/user/login	Login page	Login page	Pass
2	Enter invalid credentials	Email = 'demo@gmail.com', password = 'Demo'	Password is incorrect.	Password is incorrect.	Pass
3	Enter valid credentials	Email = 'demo@gmail.com', password = 'Demo@123'	Login successful	Login success Pass and Redirected to Dashboard	Pass

Table 4. 2 test case for login

Test cases for Logout:

T08: Check user clicks logout button

ID	Description	Expected Results	Actual Results	Pass/Fail
T08	Check user clicks logout button	User logged out of application.	Redirected to homepage	Pass

Table 4. 3 test case for logout

Pre-condition: User is already logged in.

Post-condition: After successful logout, user session is clear

sn	Test Name	Input	Excepected outcome	Acutal Results	Pass/Fails
1	Add resume	http://localhost/api/v1/resume/uplaod	Add resume page	Add resume page	pass

2	Add invalid credentials	Resume:'''	resume is required	resume is required	pass
3	Add valid credentials	Resume:''cv.png''	resume Added successfully	resume added successfully redirecting to recommendation	pass

Table 4. 4 test case for upload resume

s n	Test Name	Input	Excepte d outcome	Acutal Results	Pass/Fail s
1	delete job (method:delete)	http://localhost/api/v1/job/:id	delete job page	delete job page	pass
2	update invalid credentials	http://localhost/api/v1/job	id is required	id is required	pass
3	Add valid credentials	http://localhost/api/v1/jsalhabkclk	Location deleted	Locatio n deleted	pass

Table 4. 5 Test case for delete job

s n	Test Name	Input	Excepted outcome	Acutal Results	Pass/Fail s
1	create job	http://localhost/api/v1/job/post	recommendatio n location page	create job page	pass
2	update invalid credential s	Title:"frontend Developer", Description:"This is for demo", Salary:20000, jobType:"fulltime", experience:"entry-level", position:1, location:"demo", companyID:"dsjljfh21234"	Requirement is required	requiremen t is required	pass
3	Add valid credential s	Title:"frontend Developer", Description:"This is for demo", Requirement:"html,css,js", Salary:20000, jobType:"fulltime", experience:"entry-level", position:1, location:"demo", companyID:"dsjljfh21234"	job created successfully	create job successfull y	pass

Table 4. 6 test case for the creating job

sn	Test name	Input	Expected outcome	Actual results	Pass/fail
1	Get all jobs	http://localhost/api/v1/job/	Get all job	Get all jobs	pass
2	Invalid credentials	http://localhost/api/v1/jobs/	404 not found	404 not found	pass

Table 4. 7 test case for the get all jobs

S n	Test name	Input	Expected outcome	Actual result	Pass/f ail
1	Apply job	http://localhost:8081/api/v1/application/app/:id	Apply job route	Apply job route	pass
2	Invalid credentials	http://localhost:8081/api/v1/application/app/	JobID is missing	jobId is missing	pass
3	Invalid credentials (if user is not login)	http://localhost:8081/api/v1/application/app/kasdka1231b	User is not authenticated	User is not authenticated	pass

Table 4. 8 test case for apply jobs

sn	Test name	Input	Expected outcome	Actual result	Pass/fail
1	Open create company route	http:localhost:8081/api/v1/company/	Create company page	Create company route	pass
2	Invalid credentials	companyName:'''	Company name is required	Company name is required	pass

3	Valid credentials	companyName:"demo"	Company name created successfully	Company name created successssfully	pass
---	-------------------	--------------------	-----------------------------------	-------------------------------------	------

Table 4. 9 Test case for create company

Sn	Test name	Input	Expected outcome	actual result	Pas/fail
1	Delete company route	http://localhost:8081/ company/delete/:id	Delete company route	Delete company route	pass
2	Invalid credentials	http://localhost:8081/ company/delete/	Provide the company id	Provide the company id	pass
3	Valid credentials	http://localhost:8081/ company/delete/:abc	Company deleted successfully	Company delete successfully	pass

Table 4. 10 Test case for delete company

CHAPTER 5

CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Conclusion

The developed Job Portal system successfully addresses the challenges of traditional recruitment methods by providing a centralized, dynamic, and user-friendly web-based platform. It enables employers to post and manage job listings while allowing job seekers to register, search, and apply for jobs efficiently. The system ensures secure user authentication, facilitates application tracking, and provides an organized environment for both parties. Through its implementation, the portal streamlines the hiring process, reduces manual effort, and enhances accessibility, making it a valuable solution in the Nepali job market.

5.2 Lessons Learned

Throughout the development of the Job portal, several key lessons emerged that shaped the project's evolution. First, prioritizing user-centric design proved essential—optimizing navigation and leveraging Vite for fast load times directly boosted user engagement and satisfaction. Second, while MongoDB's flexibility accelerated development, we learned that data consistency demands careful schema planning, particularly in models like `user.model.js`, `resume.model.js`, `savedJobs.model.js`, `job.model.js`, `application.model.js`, `recommendation.model.js`, `notification.model.js`, `token.model.js`, and `company.model.js` to prevent inconsistencies during scaling. Third, testing revealed that real-time features could significantly enhance interactivity, suggesting untapped potential for future iterations. Finally, implementing robust security measures—like JWT authentication in middleware—became non-negotiable to safeguard user data and maintain trust. These insights underscore the balance between agility and rigor required for successful web applications.

5.3 Future Recommendations

To further enhance the system and provide a more comprehensive recruitment solution, the following improvements are recommended:

- **AI-Powered Job Recommendations** – Integrate advanced machine learning models to analyze resumes and job descriptions for personalized job suggestions.
- **Video Interview Module** – Allow employers to schedule and conduct video interviews directly through the platform.

- **Skill-Based Multiple Profiles** – Enable job seekers to create and manage multiple profiles within a single account.
- **Application Analytics for Employers** – Provide employers with statistics on application volume, candidate skill match percentage, and hiring trends.

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Appendix: System Screenshots

Database Overview

User table

```
_id: ObjectId('67fa5b788c56ac60591758d2')
fullname: "Rajeshan Maharjan"
email: "vijafeh241@clubemp.com"
password: "$2a$10$F/zyQjXpXa6pSqr.luh0WOSFnTue1qKHdTw96DWYTF04YXedW5zwe"
role: "worker"
failedAttempts: 0
lockUntil: null
__v: 1
resume: ObjectId('682dfafae4dde5c3c31c5682')
isVerified: false
profilePic: "profilePic-1749017745020-366837593.png"
```

SavedJob table

```
_id: ObjectId('68445022e7b10cb80bc6aa1b')
user: ObjectId('67fa5b788c56ac60591758d2')
job: ObjectId('682b131096fca21fb210d980')
createdAt: 2025-06-07T14:43:46.188+00:00
updatedAt: 2025-06-07T14:43:46.188+00:00
__v: 0
```

```
_id: ObjectId('6881185f58d61cccf7ac385e')
user: ObjectId('67fa5b788c56ac60591758d2')
job: ObjectId('6839bfa63c99809e0014140b')
createdAt: 2025-07-23T17:14:07.538+00:00
updatedAt: 2025-07-23T17:14:07.538+00:00
__v: 0
```


Resume table

```
_id: ObjectId('682da13eb50eac1f67273a2f')
user : ObjectId('67fa5b788c56ac60591758d2')
filename : "resume-1747820862510-976670179.docx"
content : "RAJESHAN MAHARJAN"
```

Frontend Developer

```
Siddhipur, lalitpur | hello@gma..."
name : "Rajeshan Maharjan"
email : "hello@gmail.com"
phone : "+977 974188756"
website : "www.rajeshanmaharjan.com.np"
location : "Siddhipur"
▸ jobTitles : Array (1)
▸ skills : Array (22)
▸ experience : Array (2)
▸ projects : Array (2)
uploadedAt : 2025-05-21T09:47:42.602+00:00
__v : 0
```

Recommendation table

```
_id: ObjectId('68458bcc37b292a871cc3d50')
user : ObjectId('68458bab37b292a871cc3d43')
resume : ObjectId('68458bcb37b292a871cc3d4a')
job : ObjectId('682b131096fca21fb210d980')
score : 70
matchingReason : "Your experience as backend developer matches this position"
notified : false
createdAt : 2025-06-08T13:10:36.597+00:00
__v : 0
```

Notification table

```
_id: ObjectId('688b2f11b4c506fd10f0711b')
user : ObjectId('67fa5b788c56ac60591758d2')
message : "Your password was changed successfully."
isRead : true
createdAt : 2025-07-31T08:53:37.618+00:00
__v : 0
```

Job table

```
_id: ObjectId('688ca15f505fbfbdc0157cd8')
title : "Frontend Developer"
description : "We are looking for a skilled Frontend Developer to join our team."
▸ requirement : Array (5)
salary : 800000
experienceLevel : "Mid Level"
location : "Kathmandu"
type : "Full-time"
position : "2"
company : ObjectId('688becbe00694531f71dfa48')
created_by : ObjectId('681dbe4ed4aafa74dfe6560c')
▸ applications : Array (empty)
__v : 0
createdAt : 2025-08-01T11:13:35.916+00:00
updatedAt : 2025-08-01T11:13:35.916+00:00
```

Company table

```
_id: ObjectId('682b0bb2b1444579ae4e33c5')
name : "Maharjan IT PVT LTD"
userId : ObjectId('6829e501d4cfc3bdebaed4ab')
createdAt : 2025-05-19T10:45:06.814+00:00
updatedAt : 2025-05-19T10:45:06.814+00:00
__v : 0
```

application table

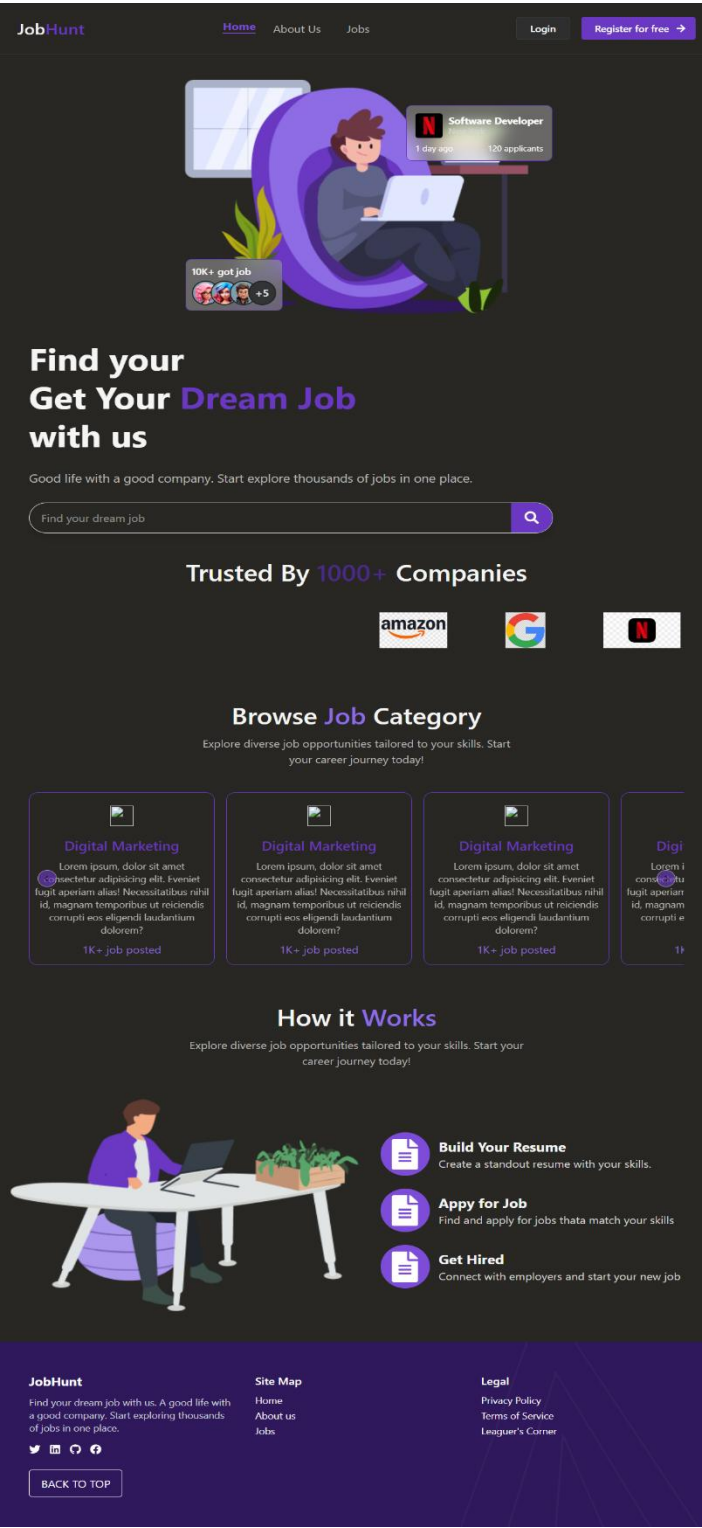
```
_id: ObjectId('682b15fb96fca21fb210d98a')  
job: ObjectId('682b131096fca21fb210d980')  
applicant: ObjectId('681dbe4ed4aafa74dfe6560c')  
status: "inProgress"  
createdAt: 2025-05-19T11:28:59.766+00:00  
updatedAt: 2025-05-19T11:37:40.279+00:00  
__v: 0
```

Token table

```
_id: ObjectId('6893e38899611b8e99c5519e')  
token: "cbc631ceade6b46bfc6f5c0acd59faec"  
otp: "458424"  
user: ObjectId('6829e501d4cfc3bdebaed4ab')  
verified: false  
createdAt: 2025-08-06T23:01:05.788+00:00  
__v: 0
```

Frontend overview

Landing page



Job page

**UI/UX Designer**
Nester IT PVT LTD

FULL-TIME SIDDHIPUR MID LEVEL

Creative UI/UX Designer needed to design intuitive user interfaces.

NPR 7.5 LPA [View Details](#)

**Frontend Developer**
Treeleaf IT PVT LTD

FULL-TIME SIDDHIPUR, LALITPUR MID LEVEL

We are looking for a skilled Frontend Developer to join our team.

NPR 8.0 LPA [View Details](#)

**Backend Developer**
Treeleaf IT PVT LTD

FULL-TIME SIDDHIPUR, LALITPUR MID LEVEL

Looking for an experienced Backend Developer to build scalable APIs.

NPR 8.5 LPA [View Details](#)

**UI/UX Designer**
Treeleaf IT PVT LTD

FULL-TIME SIDDHIPUR, LALITPUR MID LEVEL

Creative UI/UX Designer needed to design intuitive user interfaces.

NPR 7.5 LPA [View Details](#)

**UI/UX Designer**
Maharjan IT PVT LTD

FULL-TIME KATHMANDU MID LEVEL

Creative UI/UX Designer needed to design intuitive user interfaces.

NPR 7.5 LPA [View Details](#)

**Frontend Developer**
Nester IT PVT LTD

FULL-TIME SIDDHIPUR MID LEVEL

We are looking for a skilled Frontend Developer to join our team.

NPR 8.0 LPA [View Details](#)

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Maharjan IT PVT LTD

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Looking for an experienced Backend Developer to build scalable APIs.

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NPR 8.5 LPA [View Details](#)

**Frontend Developer**
Maharjan IT PVT LTD

FULL-TIME KATHMANDU MID LEVEL

We are looking for a skilled Frontend Developer to join our team.

NPR 8.0 LPA [View Details](#)

Login page

Welcome Back

RECRUITER

WORKER

Email

Enter your email

Password

Enter your password

Forgot password?

Submit

Don't have an account? [Register](#)

Register Page

Register for free

RECRUITERWORKER

Full Name

Enter your full name...

Email

Enter your email...

Phone Number

Enter your phone number...

Password

Enter password

Submit

Don't have an account? [Login](#)

Forget password

Forgot password?

No worries, we'll send you reset instructions.

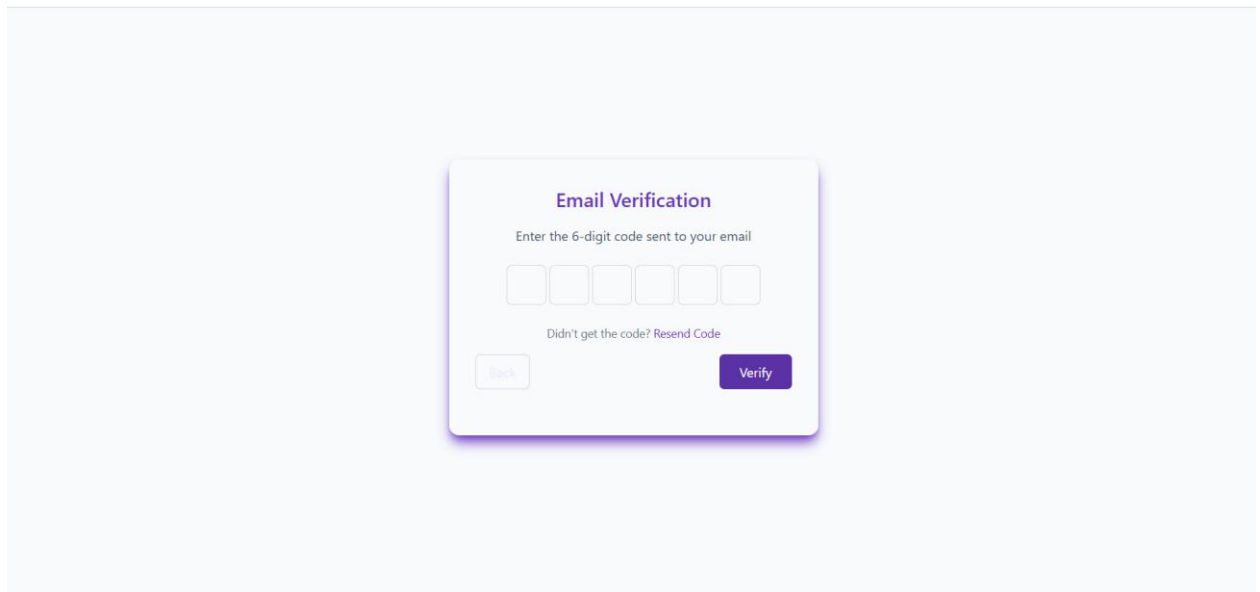
Email

Enter your email

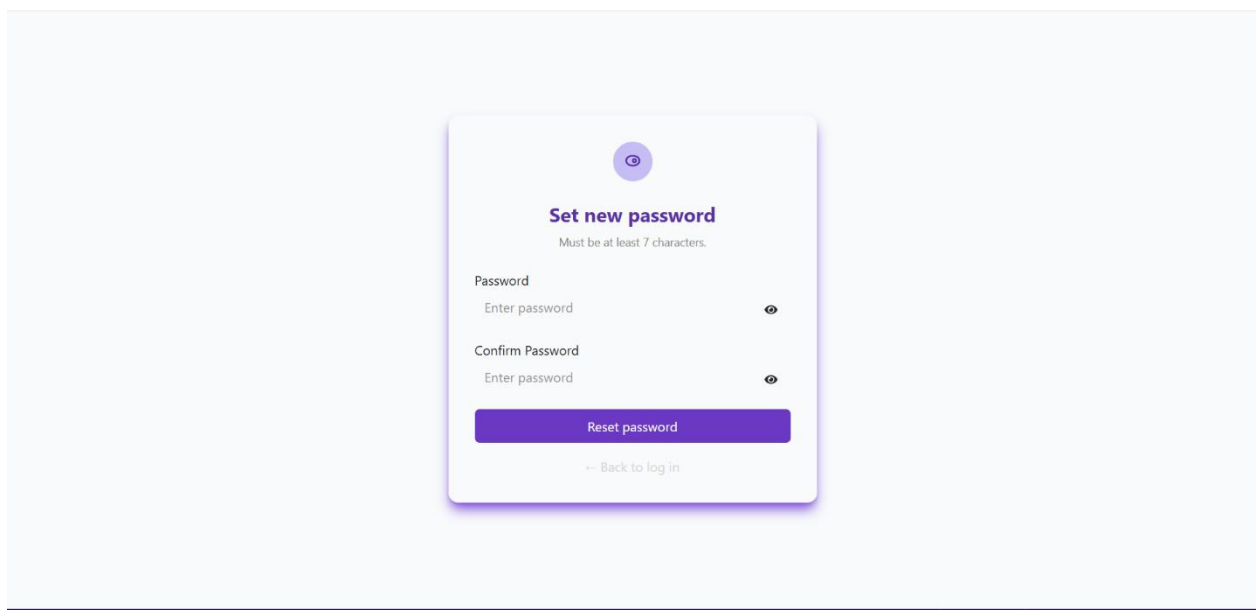
Reset password

[← Back to log in](#)

Verification code



Reset password



After login for job seeker

JobHunt

[Home](#) [About Us](#) [Jobs](#) [Recommendations](#)

 **Rajeshan Maharjan**  

Find your Get Your **Dream Job** with us

Good life with a good company. Start explore thousands of jobs in one place.



Reset password

← Back

Reset your password
Must be different to previous passwords.

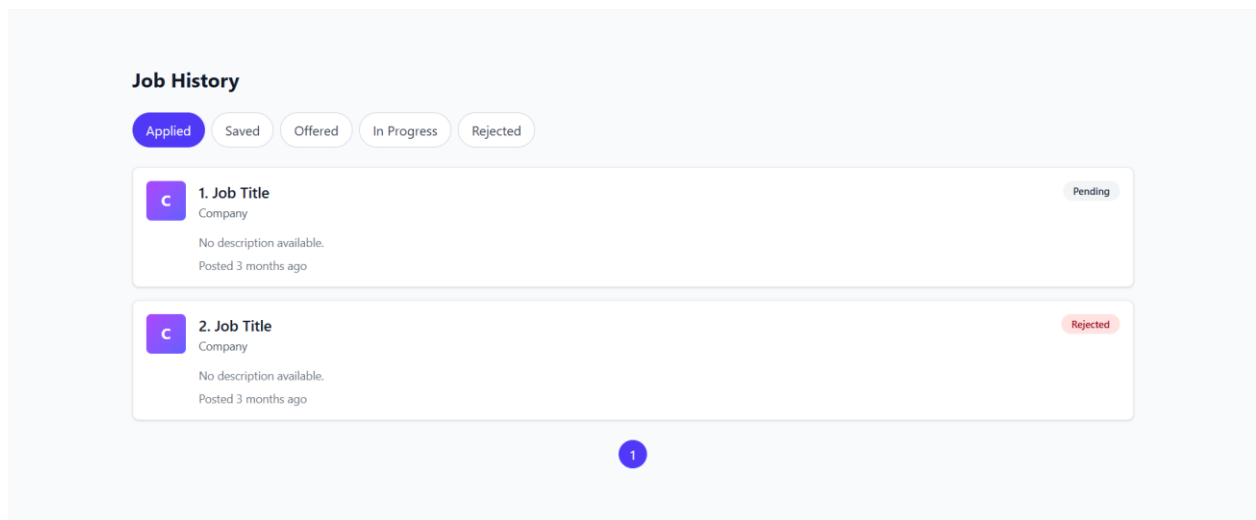
Current Password

New Password

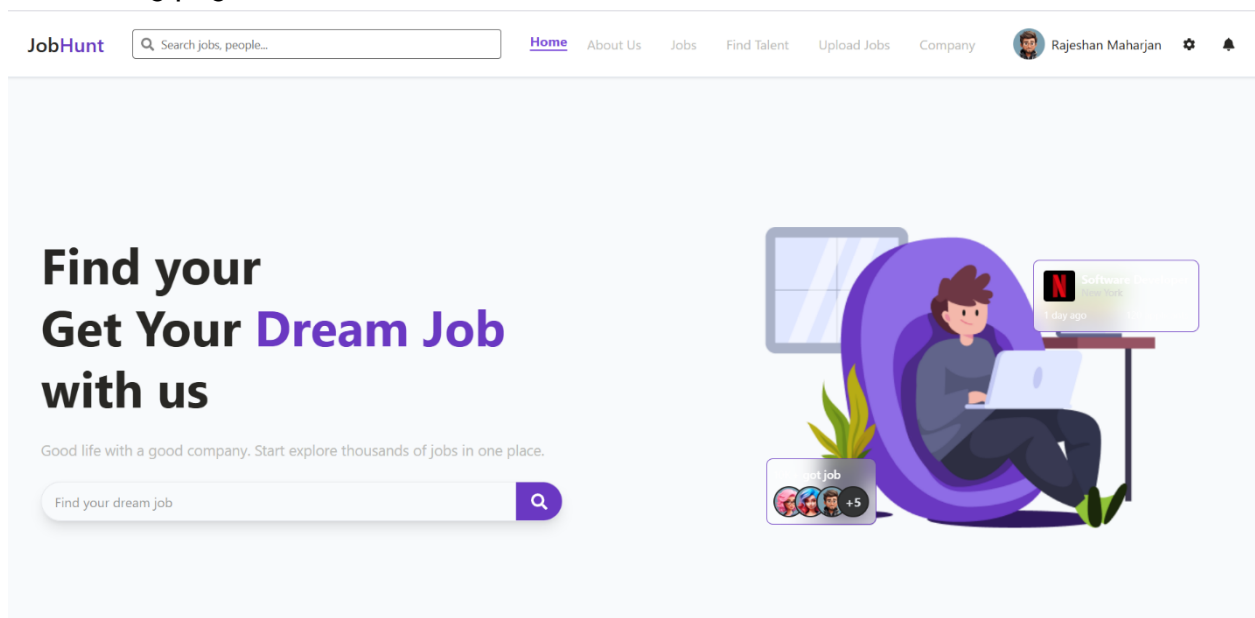
Confirm New Password

Update Password

Job history



Landing page for the recruiter



Post new job

Post a New Job

Select a Company

Select Job Type

Select Experience Level

Job Description

Post Job

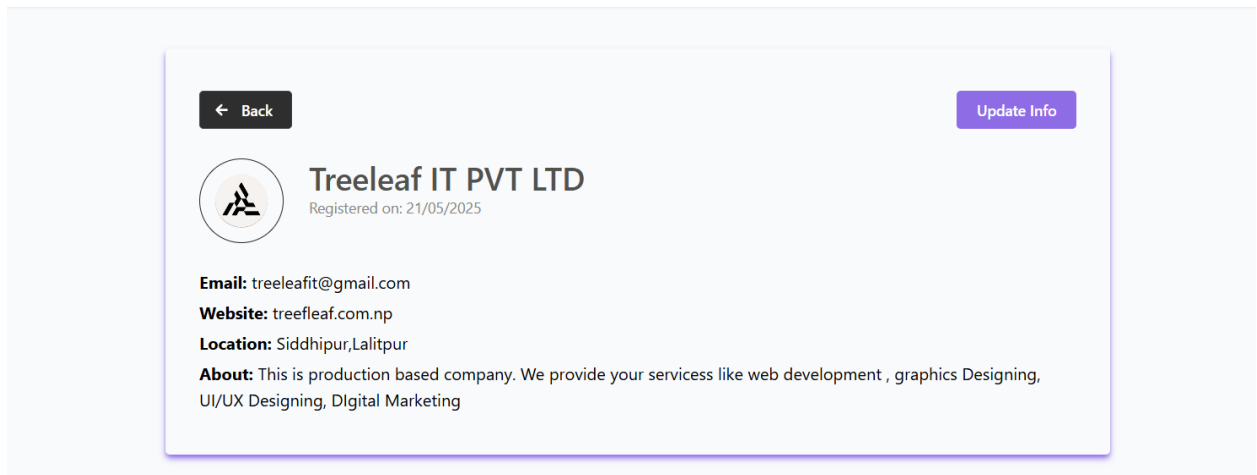
Create company and list of company

New Company

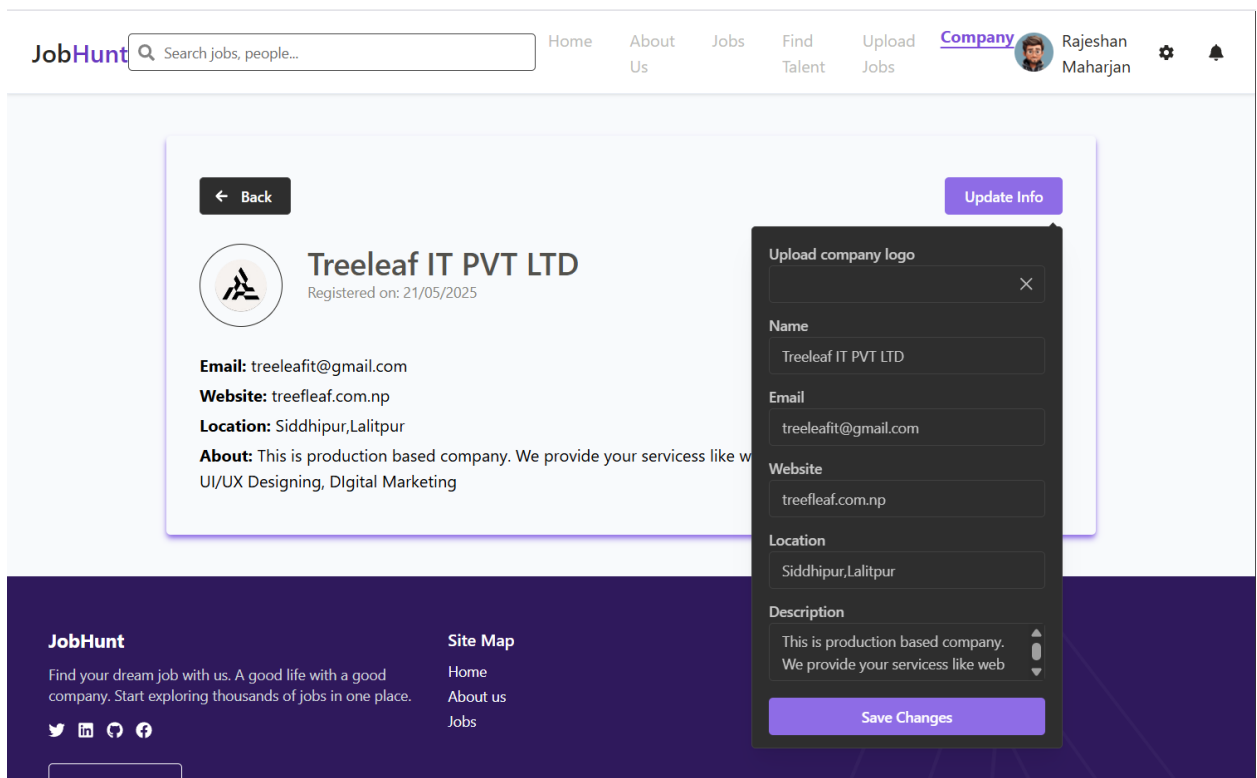
Recent Registered Companies

LOGO	NAME	REGISTERED DATE	ACTION
	name	01/08/2025	View
	Ramilo Tech PVT.LTD	23/05/2025	View
	Treeleaf IT PVT LTD	21/05/2025	View
	Nester IT PVT LTD	19/05/2025	View

Company information



Update company



Application Management

Manage and review job applications

Search applicants...

All Applications

All Jobs

Clear Filters

Rajeshan Maharjan

vijafeh241@clubemp.com

APPLIED

View

Accept

Reject

Applied for: Frontend Developer

Company: Nester IT PVT LTD • Applied on: Aug 7, 2025

Rajeshan Maharjan

vijafeh241@clubemp.com

APPLIED

View

Accept

Reject

Applied for: Frontend Developer

Company: Treeleaf IT PVT LTD • Applied on: Aug 7, 2025

Application Summary

2

Total

2

Applied

0

In Progress

0

Offered

0

Rejected